

1 Operation Object

Definition

The operation or procedure is one step within a work flow during which a manufactured quantity of an order's article is produced.

In addition to the operation number, the information listed below is needed to identify the operation:

- A written description of the work that needs to be performed
- The required workplace or the required group of identical workplaces
- In some cases, any other resources needed (e.g. tools, drawings, NC programs)
- Time needed to carry out the work (e.g. setup time, processing time)
- Target quantity (batch size)

If different machines or groups are required for production, this is what we refer to as multilevel production. Therefore, multilevel production includes several machine-related operations, which normally run one after the next. The number of operations needed for an order is not limited.

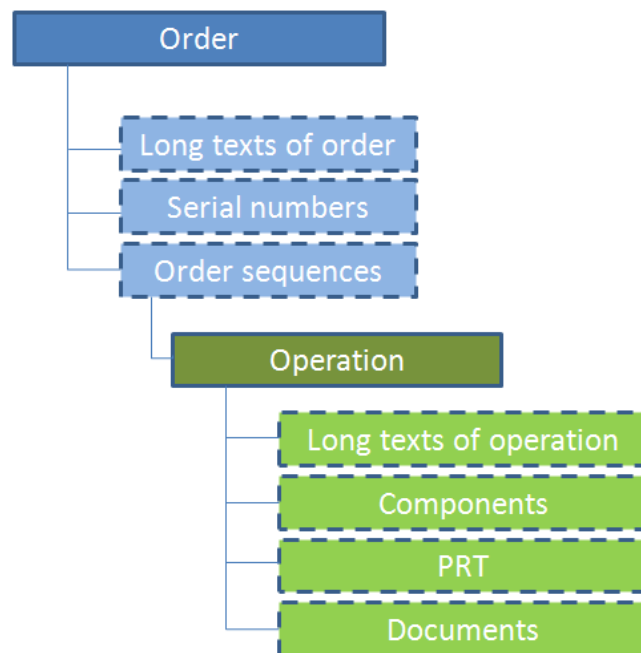
Terms used synonymously for the term operation are: procedure or order sequence/maintenance sequence (AFO). Oftentimes, the term order itself is also used synonymously.

Usage

All activities that a person carries out on a machine/work station are order and/or operation related. The posting of the order and operation answers the question what is being done and/or what activity is being carried out.

Structure

Each operation can be identified by the relevant combination of the unique order number and the sequence and operation number. This is either provided and administered by an upstream system (generally ERP system) or by the MES system itself. The object "operation" is subordinate to the object "order" and "sequence" and is structured as follows:



Please note: the object order sequence is only used if specifically requested.

Integration

The operation outputs a material with a specific material type. The operation also includes as additional information the bill of materials or rather the component list showing the materials that are needed or that are relevant in manufacturing the article. The same applies to the range of different production resources (e.g. tools) itemized in the production resources and tools list.